Agent Name: Geospatial Supply Chain Agent
What it does:
The Geospatial Supply Chain Intelligence Agent leverages a robust Supply Chain Knowledge Graph to deliver context-aware recommendations across suppliers, products and locations.
By connecting and analyzing relationships between suppliers, products and locations the agent enables to make faster and smarter supply chain decisions.
The agent helps to identify alternate suppliers based on product availability, pricing, and regional proximity.
The most challenging part in the supply chain is to search the nearest location for alternative suppliers. This agent solves the issues using geospatial co-ordinates to find the suppliers.

Key capabilities include:
Alternate supplier recommendation based on availability, location, and supplier relationships
Nearest supplier discovery using geospatial graph queries
Alternate suppliers for similar products

Dataset and why a graph fits:
Created a Supply Chain Dataset with 8000 rows having the below columns

Column	Description
Supplier_ID	Unique codes like SUP83421
Supplier_Name	Tech distributor names (e.g. Nova Components, Apex Distributors)
Product	48 distinct computer spare types (RAM, SSDs, CPUs, GPUs, laptop parts, peripherals, etc.)
Description	Detailed product specs per item
Price	pricing per product category
City	30 Major Indian cities (Mumbai, Bengaluru, Delhi, Hyderabad, Chennai, etc.)
Latitude	City coordinates
Longitude	City coordinates

A graph database is an ideal fit for supply chain systems because supply chains are naturally built around interconnected relationships between suppliers, products and locations.
Supply Chains Are Highly Connected Networks

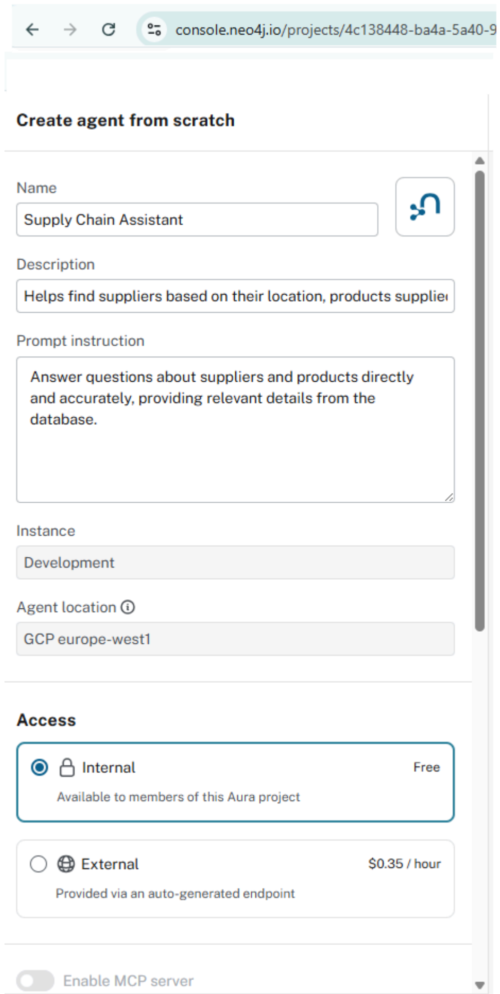
A single product may involve:
Multiple suppliers
Different transportation routes
Graph databases are designed specifically to manage and analyze these interconnected ecosystems efficiently.
Faster Alternate Supplier Discovery
When a supplier disruption occurs, graph queries can instantly identify:
Alternate suppliers
Nearby suppliers
Suppliers providing similar products
Suppliers connected through existing logistics routes
Instead of performing multiple complex table joins, the graph traverses relationships directly.

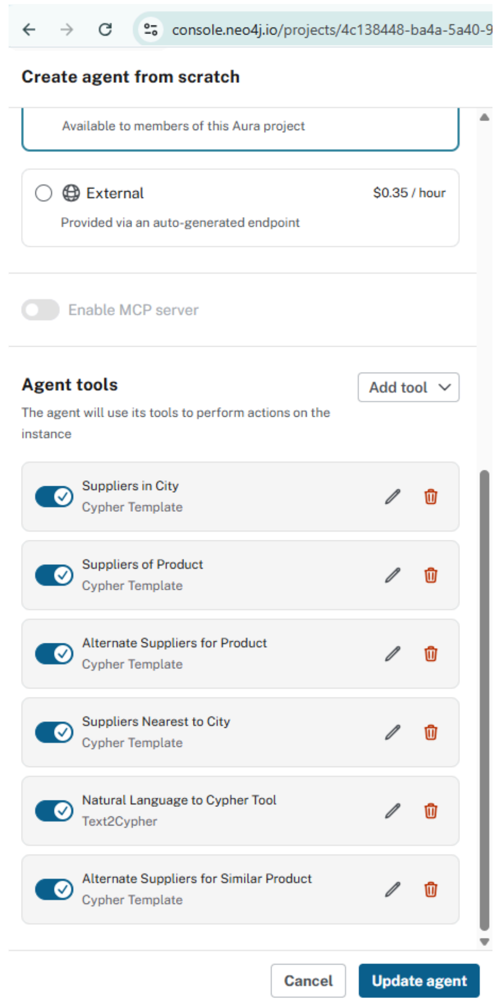
Geospatial and Nearest Supplier Intelligence
Graph databases can combine relationship intelligence with spatial data to:
Find the nearest supplier
Optimize delivery routes

Loaded the supply_chain_dataset.csv file from Github into Neo4j Aura using the below script:

```
LOAD CSV WITH HEADERS FROM <Github Path>
 AS row
CALL(row) {
MERGE (sup:Supplier {supplierId: trim(row.Supplier_ID)}) SET sup.name=row.Supplier_Name
MERGE (prd:Product {name: trim(row.Product)}) SET
prd.description=row.Description,prd.price=toFloat(row.Price)
MERGE (cy:City {name: trim(row.City)}) SET cy.location = point({latitude: toFloat(row.Latitude),
longitude: toFloat(row.Longitude)})
MERGE (sup)-[:SUPPLIES]->(prd)
MERGE (sup)-[:LOCATED_IN]->(cy)
} IN TRANSACTIONS OF 1000 ROWS
```

Agents in Aura console:

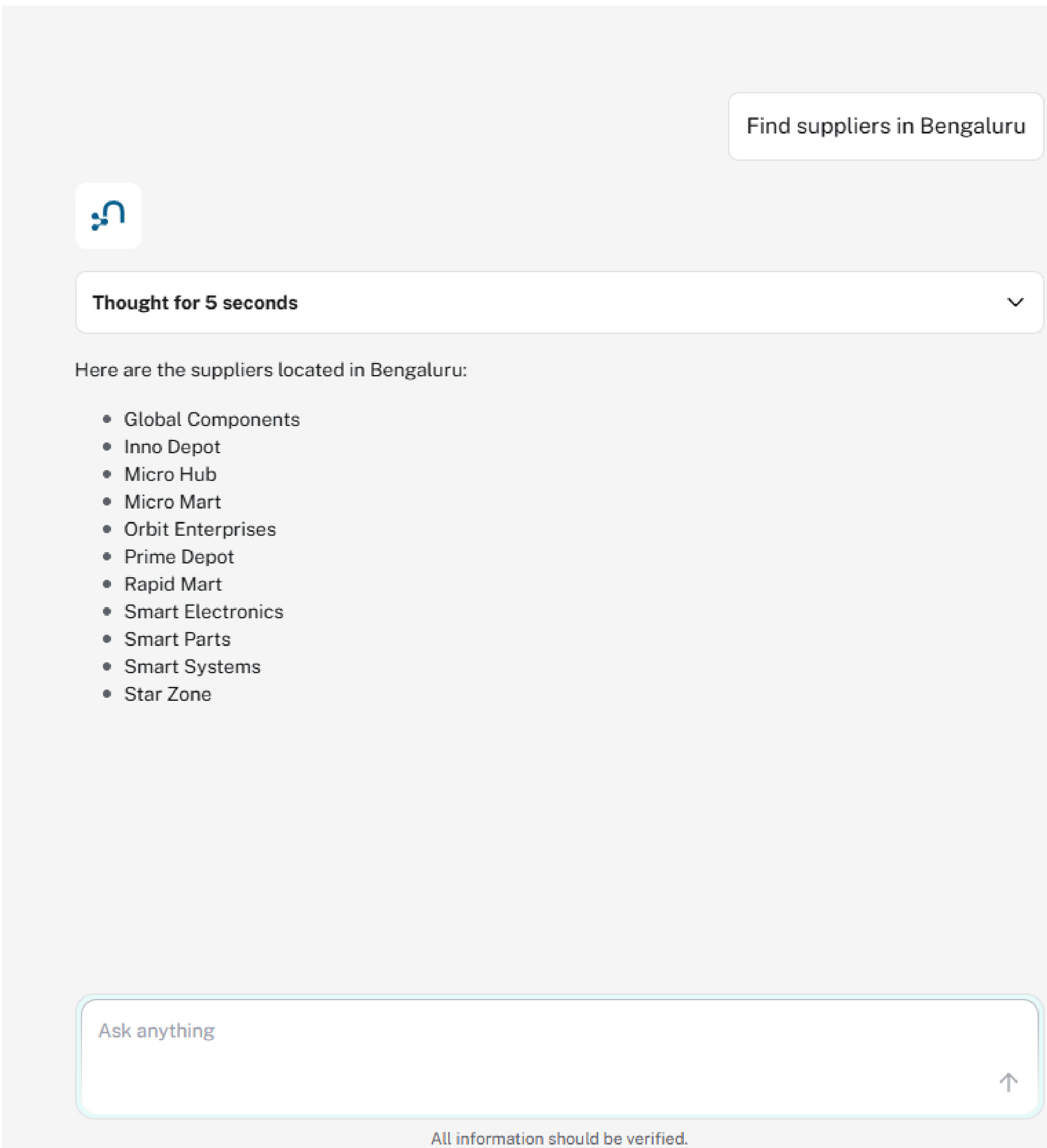




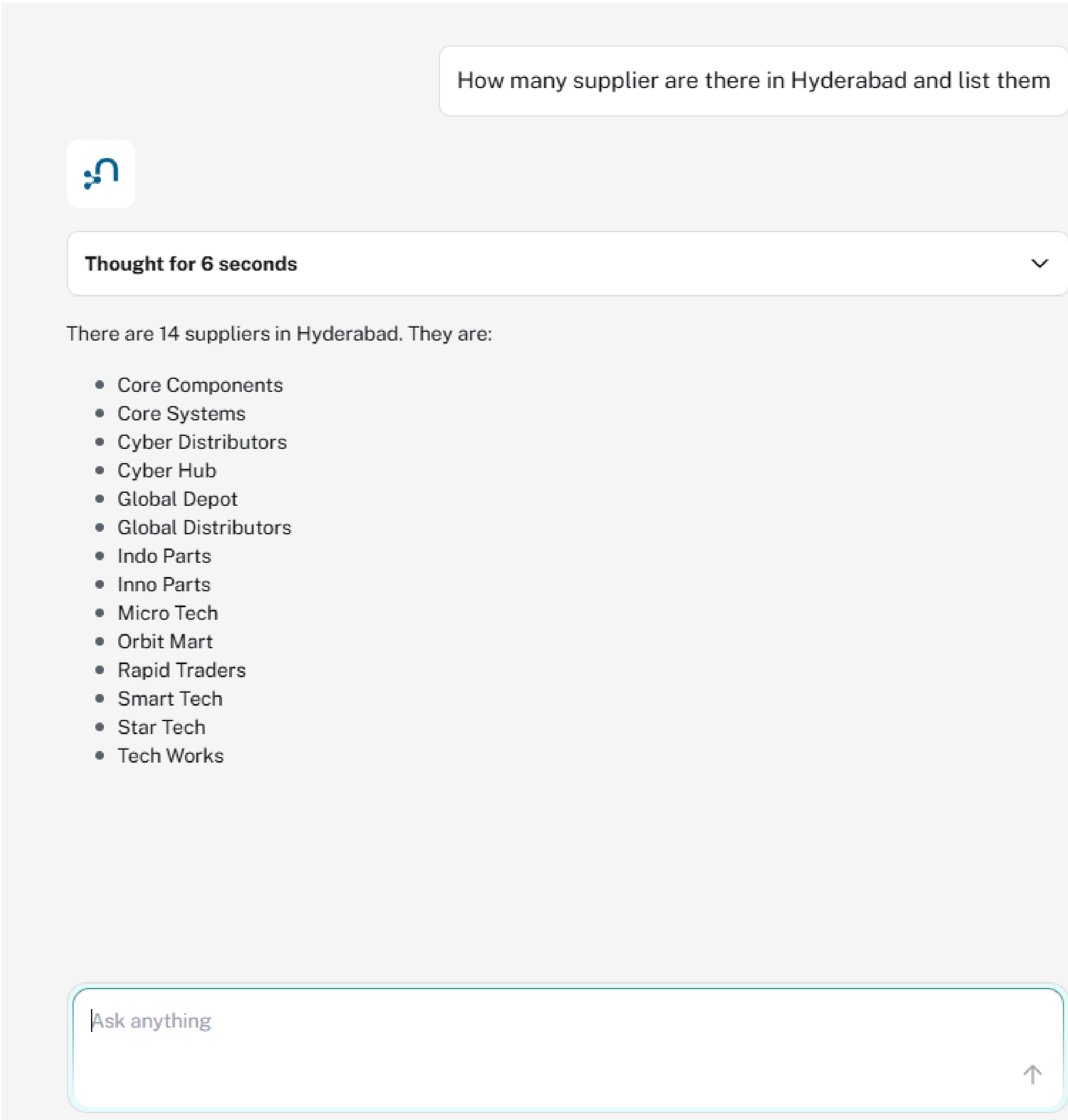
Demo of Agents in Action:

Finding suppliers in a specific City:

```
MATCH (s:Supplier)-[:LOCATED_IN]->(c:City {name: ‘Bengaluru’})
RETURN s.name as SupplierName,
s.supplierId as SupplierId,
c.name as City,
c.location as Location
ORDER BY s.name
```



Finding the no of suppliers in a specific city and listing them



Thought for 6 seconds

Reasoning

Ah, I understand. The user wants to know how many suppliers are located specifically in Hyderabad. My immediate thought is to leverage the `Suppliers_in_City` tool. This tool is designed precisely for this kind of query, allowing me to fetch a comprehensive list of all suppliers associated with a particular city. Once I have that list, the final step is straightforward: I'll simply count the number of entries in the returned data to provide the exact quantity the user is seeking.

Applying agent tool: Suppliers in City

Input

{ "cityName": "Hyderabad" }

Output

{ "keys": ["SupplierName", "SupplierId", "City", "Location"], "records": [{ "City": "Hyderabad", "Location": { "SpatialRefId": 4326, "X": 78.523714, "Y": 17.400747 }, "SupplierId": "SUP18326", "SupplierName": "Core Components" }, { "City": "Hyderabad", "Location": { "SpatialRefId": 4326, "X": 78.523714, "Y": 17.400747 }, "SupplierId": "SUP86484", "SupplierName": "Core Systems" }, { "City": "Hyderabad", "Location": { "SpatialRefId": 4326, "X": 78.523714, "Y": 17.400747 }, "SupplierId": "SUP48469", "SupplierName": "Cyber Distributors" }, { "City": "Hyderabad", "Location": { "SpatialRefId": 4326, "X": 78.523714, "Y": 17.400747 }, "SupplierId": "SUP99192", "SupplierName": "Cyber Hub" }, { "City": "Hyderabad", "Location": { "SpatialRefId": 4326, "X": 78.523714, "Y": 17.400747 }, "SupplierId": "SUP11504", "SupplierName": "Global Depot" }, { "City": "Hyderabad", "Location": { "SpatialRefId": 4326, "X": 78.523714, "Y": 17.400747 }, "SupplierId": "SUP16175", "SupplierName": "Global Distributors" }, { "City": "Hyderabad", "Location": { "SpatialRefId": 4326, "X": 78.523714, "Y": 17.400747 }, "SupplierId": "SUP38746", "SupplierName": "Indo

Finding the Suppliers of a Product & their location:

Who are the supplier of Laptop Battery 4400mAh and where are they located?

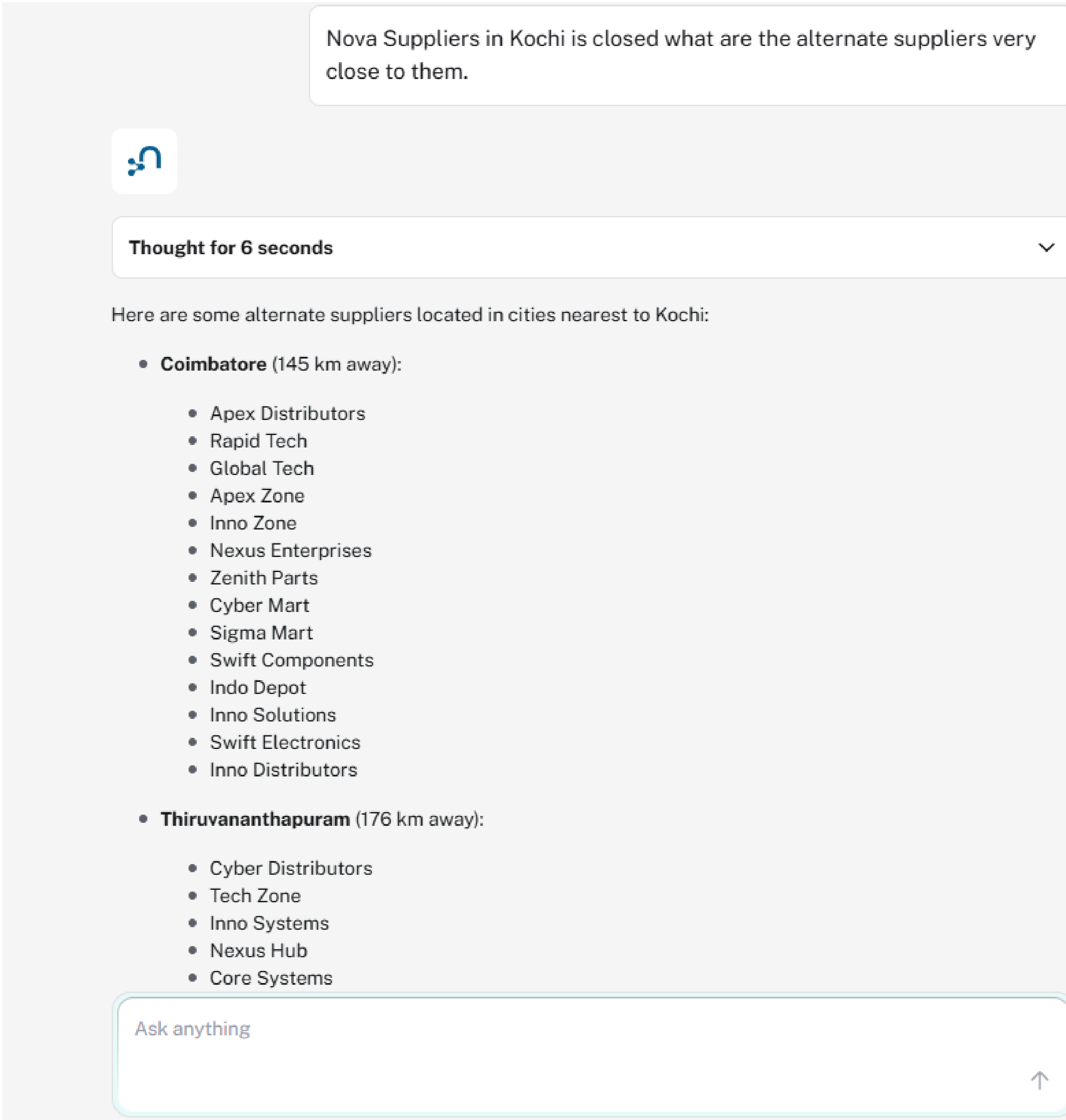
Thought for 12 seconds

Here are the suppliers of "Laptop Battery 4400mAh" and their locations:

- Zenith Electronics (Vadodara)
- Smart Systems (Bengaluru)
- Rapid Tech (Coimbatore)
- Nexus Enterprises (Visakhapatnam)
- Tech Zone (Thiruvananthapuram)
- Global Hub (Chandigarh)
- Inno Hub (Mumbai)
- Tech Parts (Visakhapatnam)
- Nexus Hub (Vadodara)
- Digital Enterprises (Ahmedabad)
- Cyber Parts (Kochi)
- Global Mart (Gurugram)
- Cyber Systems (Agra)
- Inno Systems (Nashik)
- Digital Distributors (Nashik)

Ask anything

Finding alternate nearest suppliers to a supplier which has closed in a specific location:




```
MATCH (c1:City {name: 'Kochi'})
MATCH (c2:City)
WHERE c1 <> c2
WITH c1, c2, point.distance(point({latitude: c1.latitude, longitude: c1.longitude}), point({latitude: c2.latitude, longitude: c2.longitude})) / 1000 AS distanceKm
ORDER BY distanceKm ASC
MATCH (s:Supplier)-[:LOCATED_IN]->(c2)
RETURN s.name AS supplierName, c2.name AS cityLocatedIn, distanceKm
```

What are Top 5 costly products supplied by a Supplier

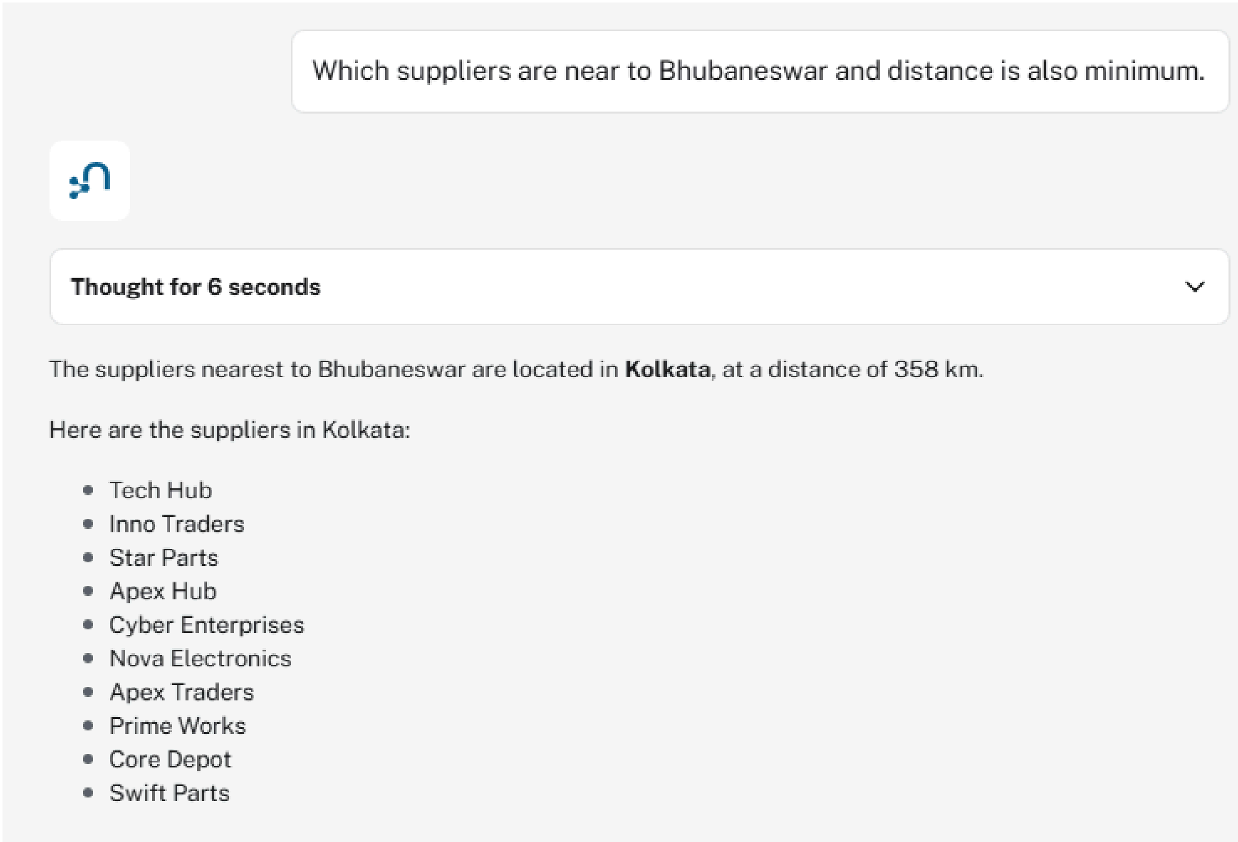
What are the top 5 high costly unique product supplied by Swift Components?

Thought for 10 seconds

The top 5 most costly unique products supplied by Swift Components are:

1. **GPU NVIDIA RTX 3060** - ₹35,882.65
2. **CPU Intel Core i7-12700** - ₹30,529.77
3. **Monitor 24" FHD** - ₹13,779.94
4. **CPU AMD Ryzen 5 5600** - ₹13,005.53
5. **RAM 16GB DDR4** - ₹6,500.00

Which suppliers are very close to a specific location and minimum:




```
MATCH (targetCity:City {name: ‘Bhubaneswar’})
WITH targetCity
MATCH (s:Supplier)-[:LOCATED_IN]->(c:City)
WHERE c <> targetCity
RETURN s.name AS supplierName, c.name AS cityLocatedIn,round(point.distance(c.location,
```

targetCity.location)/1000) AS distanceKm ORDER BY distanceKm

Thought for 6 seconds

Reasoning

My thought process begins with understanding the user's request: they're seeking suppliers located in proximity to Bhubaneswar, with a specific emphasis on finding those that are geographically closest. This immediately brings to mind a tool designed for exactly this purpose: `Suppliers_Nearest_to_City`. Therefore, my next step is to invoke this tool, making sure to specify "Bhubaneswar" as the `referenceCityName` parameter. This ensures the tool will execute its function with the correct geographical context to identify the suppliers nearest to the requested city.

Applying agent tool: Suppliers Nearest to City

Input

{ "referenceCityName": "Bhubaneswar" }

Output

{ "keys": ["supplierName", "cityLocatedIn", "distanceKm"], "records": [{ "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Tech Hub" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Inno Traders" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Star Parts" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Apex Hub" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Cyber Enterprises" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Nova Electronics" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Apex Traders" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Prime Works" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Core Depot" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Swift Parts" }, { "cityLocatedIn": "Visakhapatnam", "distanceKm": 403, "supplierName": "Tech Parts" }, { "cityLocatedIn": "Visakhapatnam", "distanceKm": 403, "supplierName": "Sigma Components" }, { "cityLocatedIn": "Visakhapatnam", "distanceKm": 403, "supplierName": "Star Solutions" }, { "cityLocatedIn": "Visakhapatnam", "distanceKm": 403,

Dashboard